



Simon Zimmermann

Software Developer

Birth June 5, 1993
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WORK EXPERIENCE

Dear Reality GmbH

2022-present

C/C++ Backend Software Developer

- Designed proprietary C-API based on DSP-prototypes using C99, C++17.
- Applied DSP-research in prototypes for internal evaluation using Python and MATLAB.
- Researched new developments in Digital Signal Processing related to Spatial audio.
- Developed custom API-wrapping tooling to facilitate rapid prototyping using C99 and Pure Data (Pd).
- Maintained DevOps pipelines using Microsoft Azure for Continuous Integration and Delivery of proprietary APIs and internal packages.

Dear Reality GmbH

2020-2022

C++ Fullstack Software Developer

- Full-stack development of real-time audio applications following test-driven development, modern C++ and agile project management principles.
- Designed cross-platform file codecs and internal tooling for maintaining data updates using FlatBuffers, Python, and Docker.
- Implemented Google Analytics and license-management for audio plugins
- Maintained self-hosted Jenkins build pipeline for Continuous Integration and Delivery.

Dear Reality GmbH

2017-2020

Researcher/QA Engineer

- Full-stack development of real-time audio applications following test-driven development, modern C++ and agile project management principles.
- Developed framework for automated real-time audio plugin testing.
- Developed DSP prototypes related to spatial audio.
- Tested real-time audio tools for quality assurance.

ISRW-Klapdor

2016-2017

Room-Acoustics Engineer

- Predictive simulation of room properties during construction to meet regulatory needs for speech intelligibility and thermal insulation in buildings.
- In-situ measurement and evaluation of room-acoustic properties during construction.

PROJECTS

dearVR Exoverb

2022

C/C++ Developer

Developed Reverb plugin with enhanced spatial perception that simulates acoustic scenes with three-dimensional depth and width using synthesized spatial room impulse responses.

Team Split Facilitation

2022

C/C++ Developer, DevOps Engineer

Transition from a single-team full-stack workflow to a frontend/backend team split with modernized DevOps infrastructure for continuous integration from an on-premise solution to a cloud-based solution.

dearVR MIX/MONITOR

2021

C++ Full Stack Developer

Developed the multichannel immersive headphone mixing plugins "dearVR MIX" and "dearVR MONITOR" which utilize headphone calibration, room simulation, and binauralization to virtualize studio environments. Supports adjustable virtual mix rooms and 26 multi-channel loudspeaker formats up to 9.1.6.

PUBLICATION

Co-Author

2021

Conference Paper

Immersive and 3D Audio: from Architecture to Automotive (I3DA): "Machine Learning-Based Room Classification for Selecting Binaural Room Impulse Responses in Augmented Reality Applications".

EDUCATION

M.Sc. Media Computer Science

2018-2020

University of Applied Sciences, Düsseldorf

Thesis title: 'Scalable Modelling of Room Acoustic Characteristics for AR devices on the Basis of Visual Information Using Deep Learning' - honors degree.

Guest Semester: M.Sc. Music Computer Science

2018

University of Music, Karlsruhe

Focus on machine learning and music information retrieval.

B.Eng. Media Engineering

2012-2017

University of Applied Sciences, Düsseldorf

Thesis title: 'A Concept for Implementing Room Acoustic Properties in the Context of a 3D-Audio Engine'.

SKILLS

Technologies

6 years:

C++17, C99, Boost, CMake, Python, gtest/gmock, Azure DevOps, Git, google-benchmark, Docker, JUCE, nvim/vim, jupyter

3 years:

conan, Pure Data (Pd), Azure-DevOps, wagtail, flask, Jenkins, Linux

1 year:

pybind11, MATLAB, Github Actions, L^AT_EX, Google-Analytics, Unreal Engine, Unity

Other

6 years:

agile workflows, Scrum/Kanban, Jira, Confluence, quality assurance

Languages

German (native), English (fluent)